

- **JRE (Java Runtime Environment):** It provides libraries, JVM, and other components to run Java applications.
- **JDK (Java Development Kit):** It includes JRE and development tools like compilers and debuggers to create Java applications.
- **JVM (Java Virtual Machine):** It interprets Java bytecode and executes it on any platform, ensuring Java's "Write Once, Run Anywhere" feature.
- **IDE (Integrated Development Environment):** A software tool that provides coding, debugging, and compiling features for Java development (e.g., BlueJ, Eclipse, IntelliJ IDEA).

Java Code:

- **Java Source Code:** Source code refers to high-level code or assembly code that is generated by humans/programmers. Source code is easy to read and modify. It is written by the programmer by using any High-Level Language or Intermediate language which is human-readable. Stored using .java files.
- **Java Object Code:** Object code refers to low-level code that is understandable by machine. In Java, Object code is generated from byte code(.class file) after going through interpreter (JVM). It is in executable machine code format. Object code contains a sequence of machine-understandable instructions that Central Processing Unit understands and executes.

Features of Java:

- **Platform Independent** (WORA - Write Once, Run Anywhere)
- **Object-Oriented**
- **Secure**
- **Robust & Reliable**
- **Platform Independent**
- **Portable**
- **Case Sensitive**
- **Java code is compiled and interpreted**